

FROM CONGRUENT NUMBERS TO BSD (LUNCH SEMINAR)

GUY SAPIRE

1. CONGRUENT NUMBERS

Let's start from one of the oldest problems in number theory.

Definition 1. A number $n \in \mathbb{N}$ is called a congruent number if there exists a right triangle with rational sides and area n .

Example 2.

- 6 is the area of a triangle with sides 3, 4, 5.
- 5 is the area of a triangle with sides $\frac{3}{2}, \frac{20}{3}, \frac{41}{6}$.
- 1 is not a congruent number using the method of descent.
(the same way we prove $\sqrt{2} \notin \mathbb{Q}$).
- Square numbers aren't congruent numbers (Fermat)

Question (Congruent Number Problem). Can you classify which numbers are congruent numbers?

Denote $C_n := \{(a, b, c) \in \mathbb{Q}^3 \mid a^2 + b^2 = c^2, \frac{ab}{2} = n\}$ the set of triangles proving n is a congruent number. We want to check if $C_n = \emptyset$ or not.

To study C_n we embed it into $\mathbb{P}^3$ as $\tilde{C}_n := \{[a : b : c : d] \in \mathbb{P}^3(\mathbb{Q}) \mid a^2 + b^2 = c^2, \frac{ab}{2} = nd^2\}$ and look at the map $\tilde{C}_n \rightarrow \mathbb{P}^1$ defined by $[a : b : c : d] \mapsto [a : b]$. Analyzing this map with tools from algebraic topology (Riemann-Hurwitz formula) we find that $\tilde{C}_n$ has genus 1. So to summarize we know three things about $\tilde{C}_n$:

- (1) It is smooth: Like in calculus or however that undergraduate course is called, the differential is regular.
- (2) It is projective: It is embedded in projective space.
- (3) It has genus 1.

Definition 3 (Elliptic Curve). An elliptic curve E over a field $\mathbb{F}$ is a smooth projective algebraic curve over $\mathbb{F}$ of genus 1 with a choice of a distinguished point O .

Using Riemann-Roch, one can show that every elliptic curve over a field of characteristic different from 2, 3 has an affine model (called the Weierstrass model) of the form:

$$E : y^2 = x^3 + ax + b$$

for some $a, b \in \mathbb{F}$ such that the polynomial in x on the right has no double roots. Here O is the point at infinity.

It is well known that E has the structure of an abelian group such that O is the identity element and the multiplication and inversion maps are morphisms of algebraic varieties. There are a few equivalent ways of defining this group structure. Some are more modern and you get the group axioms "for free". The most classical is the following:

Definition 4. Let $E/\mathbb{F}$ be an elliptic curve. We define a group structure on $E(\mathbb{F})$ with identity O (the distinguished point) by declaring that for every $P, Q, R \in E(\mathbb{F})$ rational points on the same line we have $P + Q + R = O$ (if some of P, Q, R are equal then this needs to be interpreted "with multiplicity" i.e taking the line to be tangent)

Commutativity is fairly obvious, as the definition only had to do with lines. Associativity follows from the following classical geometry claim:

Claim 5. Let $x_1, \dots, x_8$ be 8 points in the plane in "general position" (no 4 on the same line, no 7 on the same conic). Then there is a 9-th point x_9 such that every cubic curve passing through $x_1, \dots, x_8$ also passes through x_9 .

Example 6. If $\mathbb{F} = \mathbb{C}$, then E is actually a torus (genus 1 complex curve) and as a group it is isomorphic to $S^1 \times S^1$. In particular, we get that the m -torsion for every $m \in \mathbb{N}$ is $E(\mathbb{F})[m] = (\mathbb{Z}/m\mathbb{Z})^2$. This is actually true for every algebraically closed field.

Answer. So C_n is not exactly an affine model of the elliptic curve $\tilde{C}_n$. The Weierstrass model for $\tilde{C}_n$ is $E: y^2 = x^3 - n^2x$ and C_n corresponds to points with $y \neq 0$. We can see this explicitly by the following correspondence:

$$\begin{array}{ccc} C_n & \xrightarrow{\sim} & \{(x, y) \in \mathbb{Q}^2 \mid y^2 = x^3 - n^2x, y \neq 0\} \\ (a, b, c) & \mapsto & (\frac{nb}{c-a}, \frac{2n^2}{c-a}) \\ (\frac{x^2-n^2}{y}, \frac{2nx}{y}, \frac{x^2+n^2}{y}) & \leftarrow & (x, y) \end{array}$$

So now we need to investigate rational points on this curve, which is miraculously an elliptic curve. This is still an open problem today and we can use the general theory of elliptic curves to progress. For example, assuming the Birch–Swinnerton-Dyer conjecture, if $n \equiv 5, 6, 7 \pmod{8}$ then n is a congruent number.

2. MORE ON ELLIPTIC CURVES AND BIRCH–SWINNERTON-DYER

Theorem 7 (Mordell-Weil). *Given an elliptic curve E over $\mathbb{Q}$, the group of rational points $E(\mathbb{Q})$ is finitely generated.*

Definition 8 (Rank of Elliptic Curve). By Mordell-Weil, there is an isomorphism of groups $E(\mathbb{Q}) \cong \mathbb{Z}^r \times T$ where T is torsion (and finite). r is called the rank of E .

Definition 9 (Reduction mod p). Given an elliptic curve E over $\mathbb{Q}$ and p a prime number, after choosing an appropriate model of E , we can reduce the equation modulo p and get a curve over $\mathbb{F}_p$ which we denote by E_p . This curve is not necessarily an elliptic curve over $\mathbb{F}_p$, as it might be singular, but when it does happen to be an elliptic curve we say that p is a prime of good reduction.

Definition 10 (Isogeny). Given E_1 and E_2 elliptic curves over $\mathbb{F}$, an isogeny is a group homomorphism $E_1 \rightarrow E_2$ which is also a morphism of algebraic curves (that is, given locally by polynomials).

In fact, every morphism of algebraic curves $E_1 \rightarrow E_2$ that sends O_{E_1} to O_{E_2} is an isogeny.

Definition 11 (Degree of Isogeny). Every elliptic curve E over a field $\mathbb{F}$ has a field of rational functions from the points $E(\mathbb{F})$ to the field $\mathbb{F}$ (up to equivalence, if they are equal on E). This is called the function field of E and is denoted by $\mathbb{F}(E)$. An isogeny $\phi : E_1 \rightarrow E_2$ induces an embedding of the functions fields $\overline{\mathbb{F}}(E_2) \hookrightarrow \overline{\mathbb{F}}(E_1)$ by composition. The degree of ϕ , denoted $\deg(\phi)$ is the degree of this extension.

If the extension is separable then the isogeny is called separable and we can show that $\deg(\phi) = \# \ker(\phi)$.

Example 12 (p -th Frobenius Isogeny). Given an elliptic curve E over $\mathbb{F}_p$, the Frobenius isogeny $\phi : E \rightarrow E$ is defined by $\phi(x, y) = (x^p, y^p)$ in affine coordinates.

It has degree $[\mathbb{F}_p(x) : \mathbb{F}_p(x^p)] = p$ (this is not a correct explanation but gives intuition. This is actually the explanation for Frobenius on the affine line).

Theorem 13 (Hasse). *Let E be an elliptic curve over $\mathbb{F}_p$, then $|\#E(\mathbb{F}_p) - (p + 1)| \leq 2\sqrt{p}$. (This is also true over fields of order power of a prime but we don't care).*

sketch of proof. Denote by ϕ the p -th Frobenius isogeny on E . We can prove the following things:

- (1) For every $P \in E(\overline{\mathbb{F}_p})$: $P \in E(\mathbb{F}_p)$ if and only if $\phi(P) = P$, thus $E(\mathbb{F}_p) = \ker(1 - \phi)$.
- (2) $1 - \phi$ is a separable map, and so $\deg(1 - \phi) = \# \ker(1 - \phi)$. This can be seen elementarily by differentiating and seeing that the derivative is not zero.
- (3) the degree map $\text{End}(E) \rightarrow \mathbb{Z}$ is a positive definite quadratic form.
- (4) The following "Cauchy-Schwarz" style inequality is true:

Lemma 14. *Let A be an abelian group and $d : A \rightarrow \mathbb{Z}$ a positive definite quadratic form. Then $\forall a, b \in A$*

$$|d(a - b) - d(a) - d(b)| \leq 2\sqrt{d(a)d(b)}$$

Now we can rewrite the equality we are trying to prove as

$$|\deg(1 - \phi) - \deg(\phi) - \deg(1)| \leq 2\sqrt{\deg(\phi)\deg(1)}$$

And it follows exactly from the lemma.

□

Corollary 15. *With the same notations as the theorem above*

$$\#E(\mathbb{F}_p)/p = 1 + O(1/\sqrt{p}), \quad p \rightarrow \infty$$

Conjecture 16 (Birch–Swinnerton-Dyer, classical version). *Let E be an elliptic curve over $\mathbb{Q}$ with rank r , then*

$$\prod_{p \leq x} \frac{\#E(\mathbb{F}_p)}{p} \approx C \log(x)^r, \quad x \rightarrow \infty$$

For some constant C .